

crmga — CRM Build Plan

Laravel + Filament rebuild of the SuiteCRM 8.8.0 system · single-tenant first, multi-tenant SaaS in the final phase

14-week v1

2 developers

Single-tenant first → SaaS in Phase 8

Built with Claude Code

Studio per tenant

RESTful API + webhooks

Keeps n8n running

Prepared 28 July 2026 · Example schedule below assumes kickoff **Mon 3 Aug** — replace Week 1 with your real start date.

1 · What we are building

A **Laravel + Filament CRM** replacing the heavily-customised SuiteCRM 8.8.0 system. It is a **metadata-driven engine**: metadata generates the schema, screens, permissions and API, which is what makes **Studio** (customising fields, dropdowns and layouts with no deployment) possible. A modern **REST API** covers WordPress, Meta and any other platform, and a thin legacy adapter keeps the **133 existing n8n workflows running** with only a base-URL change. Each company defines its **own roles** (module × action × access level) with System Administrator and Regular User types. The core is rebuilt on a clean schema (481 messy tables → ~35 tidy ones). **Build order**: weeks 1–13 build, migrate and **go live single-tenant** for Gunness & Associates; weeks 13–14 **convert to multi-tenant SaaS** — the existing database becomes tenant #1, with no data movement.

2 · The team & who does what

Zain
BACKEND

Database & migrations · **metadata engine + SchemaManager (runtime DDL)** · models & business logic · **ACL enforcement** · **REST API + legacy adapter** · integrations · **ETL/data migration** · security, performance, deployment · **the Phase 8 multi-tenancy conversion.**

Shahmeer
FRONTEND

The entire Filament interface · **dynamic rendering from metadata** · every module's screens · activity timeline · dashboards · DNC & Hot/Warm · **role-matrix UI** · all **Studio** builder screens · settings · **the Phase 8 super-admin panel.**

Product Owner
YOU

Field-mapping answers · the open decisions (hosting, DNS/TLS, Studio governance) · priorities · UAT sign-off · the sanitized dump (locally, Phase 6) · provider credentials (Phase 5).

3 · Timeline (14 weeks) — multi-tenancy is the final phase

PHASE	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
P1 · Foundation + metadata engine		M1												
P2 · Data model + ACL + first screens					M2									
P3 · STUDIO — fields · dropdowns · layouts							M3							
P4 · CRM complete								M4						
P5 · REST API + integrations									M5					
P6 · Data migration + UAT (ETL runs LOCALLY)										M6				
P7 · Hardening + GO LIVE (single tenant)												M7		
P8 · MULTI-TENANCY conversion → SaaS													M8	

◆ M1 metadata engine · M2 schema + ACL · M3 **Studio** · M4 CRM complete · M5 REST API + integrations · M6 data migrated + UAT · M7 🚀 **GO LIVE (single tenant, wk 13)** · M8 multi-tenant SaaS.

4 · Phase detail — tasks, owners & "done when"

P1 Foundation + metadata engine

3 Aug – 14 Aug · Week 1–2 · Zain + Shahmeer

- [Zain] Scaffold Laravel 11 + Filament + spatie + Passport/Sanctum + Horizon; CI (pint/phpstan/pest)
- [Zain] **Tenancy-ready skeleton** — tenant migrations folder, settings table, central route stub, + a CI test that fails on a `tenant_id` column
- [Zain] Users + auth backend (2FA, password policy); **metadata registry** (modules/fields/option-lists/layouts/change-log)
- [Zain] **SchemaManager** — safe runtime DDL: validate → plan → snapshot → apply → log → cache bump (`_custom` sidecars)
- [Shahmeer] Filament panel shell + theme + navigation; authentication screens
- [Shahmeer] **FieldTypeRegistry** (type → component + column + cast + validation) + design-system components

M1 Done when: a field added in metadata becomes a real column, logged

P2 Data model + ACL + first screens

17 Aug – 4 Sep · Week 3–5 · Zain + Shahmeer

- [Zain] Contactable base + `HasCustomFields`; polymorphic activities + audit; `EmailAddress` morph
- [Zain] **ACL engine** — module × action × All/Owner/None, policies + shared query scopes, user types, 29 roles seeded
- [Zain] Entities: Company, Lead (verticals incl. **Study + LMIA**), Assessment, Student, Client, Affiliate, Newsletter, SMS/Calls
- [Zain] ETL transformers started early so mapping gaps surface in week 5, not week 11
- [Shahmeer] **DynamicResource** — builds table/form/detail/filters from metadata (every screen depends on it)
- [Shahmeer] Company + Lead screens (vertical-aware panels); reusable list-page framework (saved views, filters, bulk, export)

M2 Done when: Owner-level access enforced identically in UI and API

P3 STUDIO — fields · dropdowns · layouts

31 Aug – 18 Sep · Week 5–7 · Shahmeer lead · Zain

- [Shahmeer] **Field Manager** — all field types + flags, auto-generated `LBL_` label keys, delete with impact warning
- [Shahmeer] **Dropdown Editor** — lists and items, value vs label, reorder, used-by warning
- [Shahmeer] **Layout Editor** — which fields appear, order and panel, per list/detail/edit/search view; versioned + preview
- [Shahmeer] Change-history screen with rollback
- [Zain] Full field-type DDL coverage, safe type changes, soft-delete with impact check, layout versioning, change log + rollback
- *Deferred: Module Builder and Relationship Manager (post-launch)*

M3 Done when: an admin customises fields and layouts with no deploy

P4 CRM complete

14 Sep – 2 Oct · Week 7–9 · Shahmeer lead · Zain

- [Shahmeer] Remaining module screens; **Assessment scorecard**; **activity timeline** on every record
- [Shahmeer] **Dashboard** widgets (hot/warm, pipeline, my tasks, calls to make, attention-needed)
- [Shahmeer] **DNC + Hot/Warm**; **role-matrix UI** + user management; settings screens; global search
- [Zain] Settings store with encrypted secrets; scheduled jobs + daily count reports; dashboard data services; performance pass

M4 Done when: staff can run the business end to end

P5 REST API + integrations

28 Sep – 16 Oct · Week 9–11 · Zain lead · Shahmeer

- [Zain] **v1 REST** from metadata — filtering/sort/includes/cursor pagination/idempotency; UTC boundary
- [Zain] OAuth2 + personal tokens with **scopes**, rate limits, logging; ACL applied identically to the UI
- [Zain] **Thin /Api/V8/* legacy adapter** — the 133 n8n workflows keep running on a base-URL change
- [Zain] **FieldMapper** + WordPress + Meta Lead Ads + generic signed ingest; click-to-call (Asterisk, per-user extension); SMTP send
- [Shahmeer] API client/token screens, integration + field-mapping screens, API docs page, compose/call/SMS UI

M5 Done when: external apps CRUD; n8n keeps working unchanged

P6 Data migration + UAT (ETL runs LOCALLY)

12 Oct – 23 Oct · Week 11–12 · Zain lead · Shahmeer

- [Zain] `crm:migrate-legacy` — dry-run, idempotent, resumable; UTC dates, `primary_email` join, dedupe, Study/LMIA → Lead verticals
- [Zain] Import `fields_meta_data` + view defs into **Studio metadata**; reconciliation report
- [Shahmeer] Post-migration verification and fixes; run UAT sessions with staff

M6 Done when: counts reconcile to the audit; UAT signed off

P7 Hardening + GO LIVE (single tenant)

19 Oct – 30 Oct · Week 12–13 · Zain lead · Shahmeer

- [Zain] /security-review + remediation; Octane/FrankenPHP; backups with verified restore; **cutover runbook** (parallel run → switch)
- [Shahmeer] Final polish (empty/error states, accessibility, responsive); staff + admin user guides

M7 Done when: Gunness live on the new CRM, backups + rollback tested

P8 MULTI-TENANCY conversion → SaaS

26 Oct – 6 Nov · Week 13–14 · Zain lead · Shahmeer

- [Zain] Install `stanc/tenancy`; **central DB** + Tenant model; bootstrappers for DB/cache/queue/filesystem
- [Zain] Subdomain identification (wildcard DNS/TLS); **promote the live DB to tenant #1 — no data movement**
- [Zain] `tenant:create` provisioning; settings moved to per-tenant scope; isolation proven by tests
- [Shahmeer] **Super-admin panel** — companies list, create-company wizard, Studio governance + change-request queue

M8 Done when: live install runs as tenant #1; a 2nd company onboardable

5 · Technology stack

PHP 8.3 & **Laravel 11** · **Filament v3** (staff UI) · **stanc/tenancy** (database-per-tenant) · **spatie/permission** (29 roles) · **Laravel Passport** (OAuth2 API) · **MySQL/MariaDB** · **Redis + Horizon** (jobs/scheduler) · **Reverb** (click-to-call) · **FrankenPHP/Octane** (speed) · **Pest / PHPStan / Pint + GitHub Actions** (tests & CI) · plus the custom **metadata engine** (SchemaManager · FieldTypeRegistry · DynamicResource), **OpenAPI 3.1**, and a signed **webhook dispatcher**.

6 · Decisions to lock BEFORE we start

DECIDE FIRST — the plan does not fit 14 weeks with 2 developers

- Scope is already cut hard (Module Builder, relationships, webhooks, native social connectors, import wizard, report builder, workflow rewrite all deferred) and the remainder is still **101 days for Zain and 89 for Shahmeer against 70 each**.
- **Option 1 (recommended) — two releases:** weeks 1–14 deliver the CRM, permissions, all screens, REST API + legacy adapter, data migration and go-live **without Studio**; Studio + tenancy become Release 2 (~6 weeks).
- **Option 2:** keep Studio in Release 1 and move the tenancy conversion to Release 2.
- **Option 3:** add the third developer back — the full 14-week plan becomes achievable.
- **Option 4:** extend to ~19 weeks for this scope with honest dates.
- Whichever is chosen, the **metadata engine stays in Phase 1** — Studio and the dynamic API both depend on it.

BEFORE PHASE 1

- Hosting and backups (rec: Docker + FrankenPHP on a cloud VPS).
- Confirm Filament (no separate SPA) and keeping the source `char(36)` UUID primary keys (rec: yes to both).
- Confirm the reduced v1 scope.
- **Tenancy-ready rules are mandatory from day one** — all CRM migrations in `database/migrations/tenant/`, never a `tenant_id` column, per-company config in a `settings` table. A CI test enforces them.

LATER — per phase, not blocking now

- Phase 5: WordPress + Meta credentials.
- Phase 6: the sanitized dump — used **locally** by Zain.
- Phase 7: UAT sign-off; rotate the credentials that have circulated.
- **Phase 8:** the domain + **wildcard DNS/TLS** for company subdomains, the central admin domain, and the default Studio governance mode for new companies.

Full rationale + recommendations are in `docs/ARCHITECTURE.md` §5. You can reply “go with the recommendations” or change specific items.